POWER SYSTEM -1 LAB

FINAL PROJECT

EEE-308

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I. Problem Statement: Formation of bus admittance matrix, load flow study and fault analysis in a 6-bus power system

Problem description:
Consider the one-line diagram of a 6-bus power system given in Fig.1.

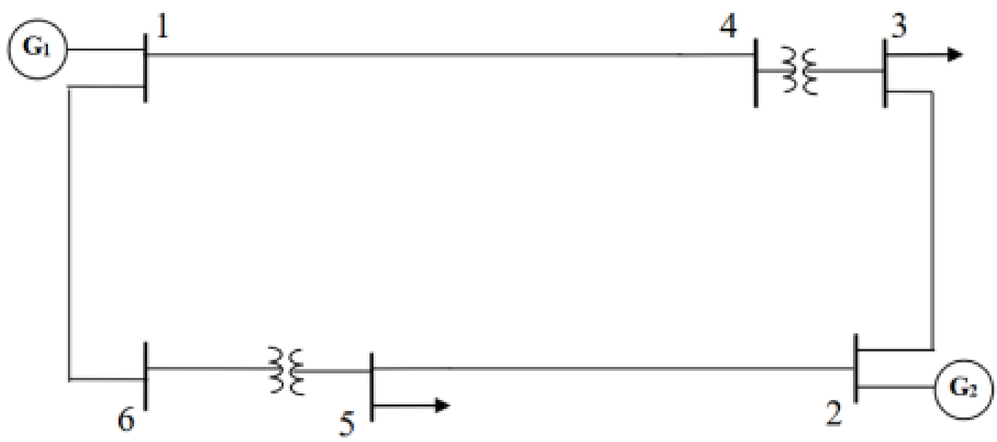


Fig. 1: One-line diagram of the network in study

Tasks:

1. **Construct** the given 6-bus system in POWER WORLD SIMULATOR and **show** the detailed diagram

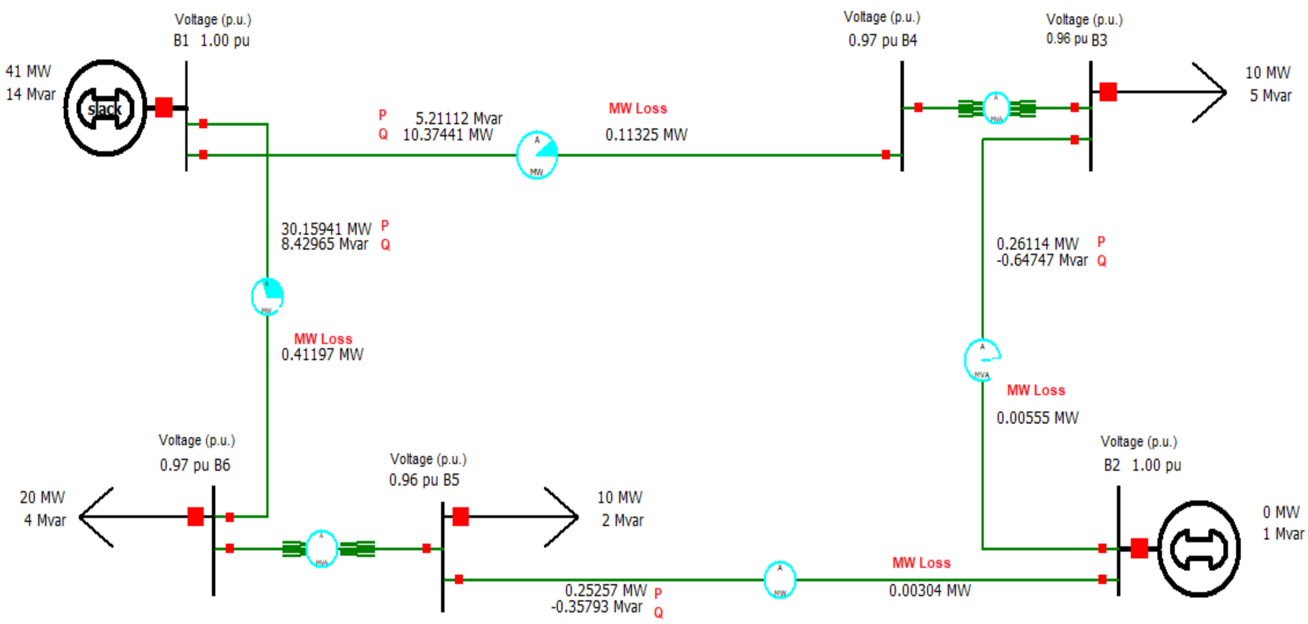


Fig. 2: POWER WORLD SIMULATOR of the 6-Bus System

2. Perform a load flow analysis of the given system and show the following in a power flow diagram:

I. Bus Voltages

ID	Voltages (pu)
B1	1.00
B2	1.00
B3	0.96
B4	0.97
B5	0.96
B6	0.97

II. Active Power flow in transmission lines

From Bus	To Bus	Active Power (MW)
1	4	10.37441
1	6	30.15941
3	2	0.26114
5	2	0.25257

III. Reactive Power flow in transmission lines

From Bus	To Bus	Reactive Power (MVAR)
1	4	5.21112
1	6	8.42965
3	2	-0.64747
5	2	-0.35793

IV. Total Power loss in transmission lines

From Bus	To Bus	Total Power Loss (MW)
1	4	0.11325
1	6	0.41197
3	2	0.00555
5	2	0.00304

V. Voltage Regulation in transmission lines between Bus-1 to Bus-4

$$\% VR = \frac{V_S - V_R}{V_R} \times 100\%$$

Bus-1 → 1.00 pu Bus-4 → 0.97 pu

$\therefore$ V_S → 1.00 pu V_R → 0.97 pu

$$\% VR = \frac{1 - 0.97}{0.97} \times 100\%$$

$$\Rightarrow \boxed{\% VR = 3.093\%}$$

$$\text{Effficeincy} = \frac{V_{OUT}}{V_{in}} \times 100\%$$

$$\Rightarrow \frac{0.97}{1.00} \times 100\%$$

$$\Rightarrow \boxed{97\%}$$

3. **Calculate** the Y-bus admittance matrix of the system in MATLAB. **Submit** the used code and show the output admittance matrix. **Verify** the result using POWER WORLD SIMULATOR.

Y-Bus Matrix from MATLAB

Editor - Ybus_matrix.m | Variables - Y_Bus

Y_Bus

6x6 complex double

	1	2	3	4	5	6	7	8
1	0.9651 - 5.7903i	0.0000 + 0.0000i	0.0000 + 0.0000i	-0.3217 + 1.9301i	0.0000 + 0.0000i	-0.6434 + 3.8602i		
2	0.0000 + 0.0000i	0.0441 - 0.2647i	-0.0257 + 0.1544i	0.0000 + 0.0000i	-0.0184 + 0.1103i	0.0000 + 0.0000i		
3	0.0000 + 0.0000i	-0.0257 + 0.1544i	0.0257 - 0.1544i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i		
4	-0.3217 + 1.9301i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.3217 - 1.9301i	0.0000 + 0.0000i	0.0000 + 0.0000i		
5	0.0000 + 0.0000i	-0.0184 + 0.1103i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0184 - 0.1103i	0.0000 + 0.0000i		
6	-0.6434 + 3.8602i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.6434 - 3.8602i		
7								
8								

Command Window

```
  Columns 1 through 5

   0.9651 - 5.7903i   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.3217 + 1.9301i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0441 - 0.2647i  -0.0257 + 0.1544i   0.0000 + 0.0000i  -0.0184 + 0.1103i
   0.0000 + 0.0000i  -0.0257 + 0.1544i   0.0257 - 0.1544i   0.0000 + 0.0000i   0.0000 + 0.0000i
  -0.3217 + 1.9301i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.3217 - 1.9301i   0.0000 + 0.0000i
   0.0000 + 0.0000i  -0.0184 + 0.1103i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0184 - 0.1103i
  -0.6434 + 3.8602i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
```

MATLAB Code for Y-Bus Matrix:

```
clc
clear
fprintf('Line Data\n\n')
fprintf('      nl')
fprintf('          nr')
fprintf('          R')
fprintf('          X')

%           FBus  TBus  R          X
%           nl    nr    pu         pu
Line_Data=[1      4     0.08402    0.50410
           1      6     0.04201    0.25205
           3      2     1.05020    6.30119
           5      2     1.47028   8.8216];


j=sqrt(-1); i=sqrt(-1);
nl=Line_Data(:,1); nr=Line_Data(:,2);
R=Line_Data(:,3); X=Line_Data(:,4);
nbr=(length(Line_Data(:,1))); % Number
of branches
nbus=max(max(nl),max(nr)); % Number of
Buses of the transmission line
Z=R+j*X; % Branch impedance
y=ones(nbr,1)./Z; % Branch admittance

%Computation of off diagonal elements
for k=1:nbr
    Y_Bus(nl(k),nr(k))=-y(k);
%This computes the elements Y12, Y13,
Y14, Y23 etc.

Y_Bus(nr(k),nl(k))=Y_Bus(nl(k),nr(k));
%Y12=Y21, Y23=Y32,etc
end

%Computation of diagonal elements
for n=1:nbus
    for k=1:nbr
        if nl(k)==n
            Y_Bus(n,n)=Y_Bus(n,n)+y(k);
        elseif nr(k)==n
            Y_Bus(n,n)=Y_Bus(n,n)+y(k);
        else, end
    end
end
fprintf('Bus Admittance Matrix\n');
Y_Bus
```

Verification of Y-Bus Matrix in POWER WORLD SIMULATOR

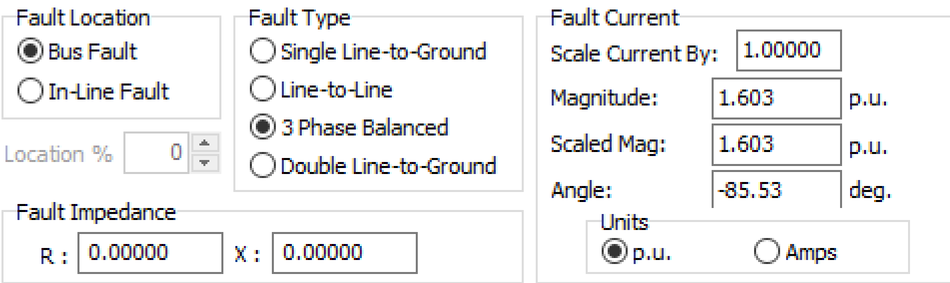
Y Bus (Bus Admittance Matrix)

	Number	Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5	Bus 6
1	1	B1	0.97 - j5.79			-0.32 + j1.93		-0.64 + j3.86
2	2	B2		0.04 - j0.26	-0.03 + j0.15		-0.02 + j0.11	
3	3	B3		-0.03 + j0.15	0.03 - j7.67	-0.00 + j7.52		
4	4	B4	-0.32 + j1.93		-0.00 + j7.52	0.32 - j9.45		
5	5	B5		-0.02 + j0.11			0.02 - j3.44	-0.00 + j3.33
6	6	B6	-0.64 + j3.86				-0.00 + j3.33	0.64 - j7.19

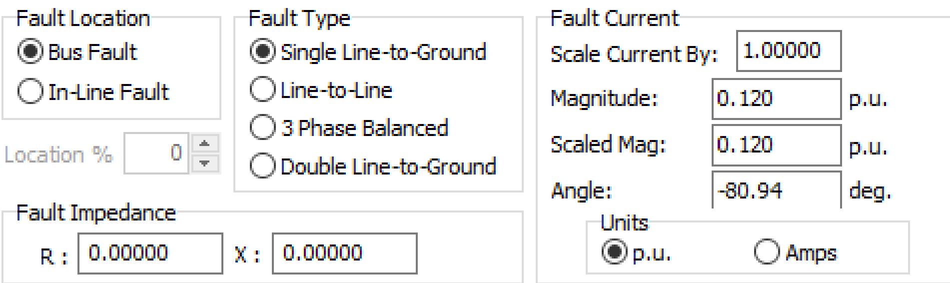
3. **Perform** fault analysis in POWER WORLD SIMULATOR and hence **determine** the fault current for balanced three-phase fault, single line-to-ground fault, line-to-line fault and double line-to-ground fault at bus 5 through a fault impedance.

I. **Z_f = 0 pu**

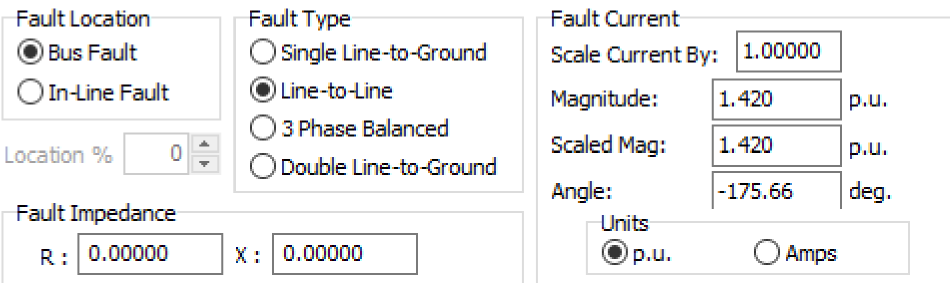
a.) Balanced Three-Phase Fault



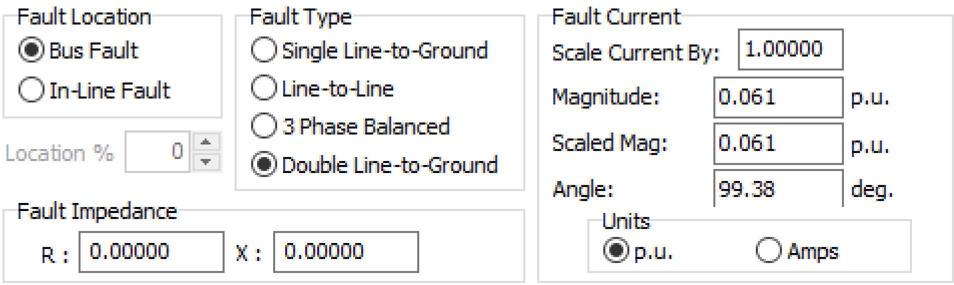
b.) Single Line-to-Ground Fault



c.) Line-to-Line Fault

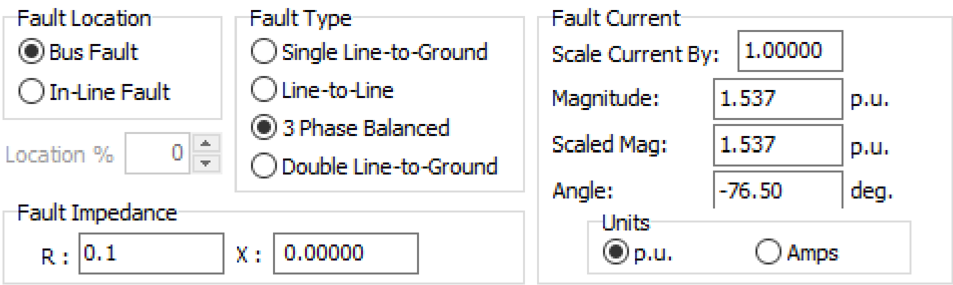


d.) Double Line-to-Ground Fault

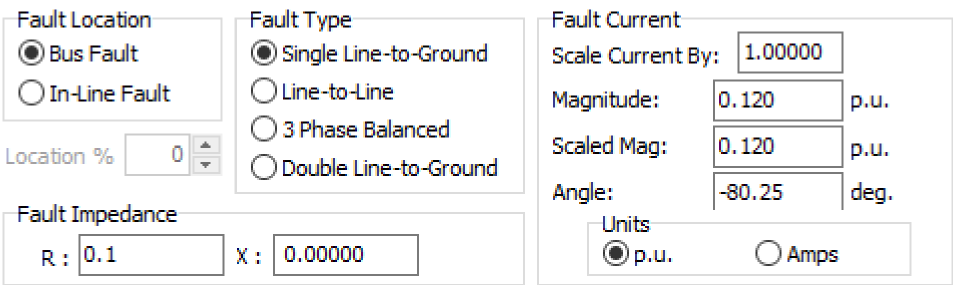


II. **Z_f = 0.1 pu**

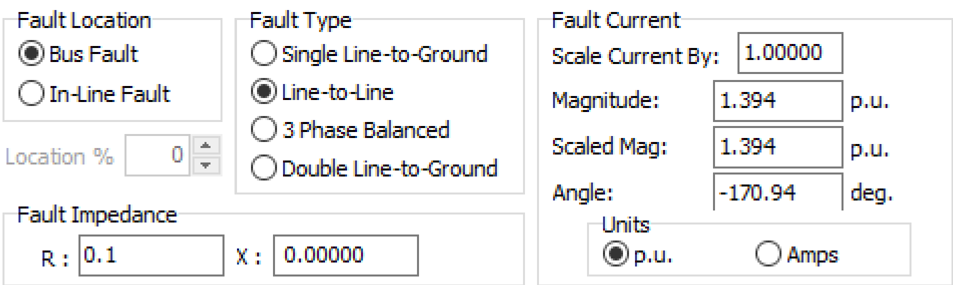
a.) Balanced Three-Phase Fault



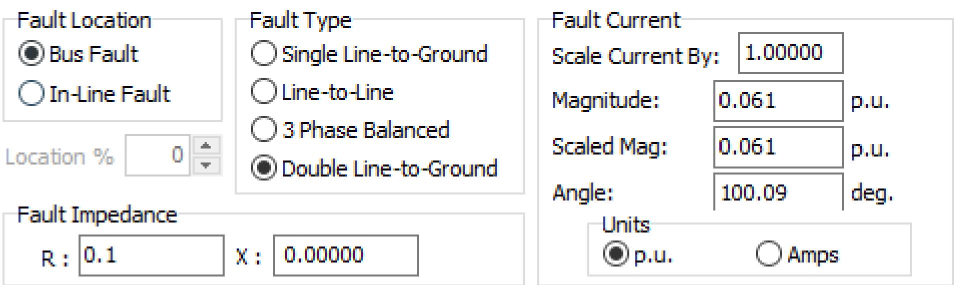
b.) Single Line-to-Ground Fault



c.) Line-to-Line Fault



d.) Double Line-to-Ground Fault



Project

Fault Analysis of the 6-Bus System

Fault Imp.	**Fault**	Magnitude (pu)	Scaled Mag (pu)	Angle
Zf=0	3-Phased Balanced	1.603	1.603	-85.53
	Single Line-to-Ground	0.120	0.120	-85.53
	Line-to-Line	1.420	1.420	-175.66
	Double Line-to-Ground	0.061	0.061	99.38
Zf=0.1	3-Phased Balanced	1.537	1.537	-76.50
	Single Line-to-Ground	0.120	0.120	-80.25
	Line-to-Line	1.394	1.394	-170.94
	Double Line-to-Ground	0.061	0.061	100.09